

Related Work:

Related Work Summary Table

Title	Dataset Name & URL	Dataset Description	Method Name	Accuracy	Pros	Cons	Citation
Microscopic Image of Different Processed Rice Varieties of Bangladesh	PRBD: Microscopic Image of Different Processed Rice Varieties of Bangladesh https://data.mendeley.com/datasets/sfp9s96prh/1	This dataset contains ten unique rice categories, which are: Aush, Beroi, BR-28, BR-29, Ghee Bhog, Katari Nazir, Katari Siddho, Swarna, Miniket, and Chinigura, in Bangladesh. These were collected from the local markets of Dhaka, Bangladesh.	DenseNet 201	Original Dataset- 93%	Regional varieties, nutritional information, global diversity, popularity data, Miniket rice, local types, undocumented quality, unverified nutritional claims, need for further research	visual similarity, limited generalization, single-source data, no external validation, controlled environment only.	Md Tahsin et al., 2025 [1]
Rice Classification Using Spatio-Spectral Deep Convolutional Neural Network	RGB and VIS/NIR Hyperspectral Imaging Data for 90 Rice Seed Varieties https://zenodo.org/records/3241923	90 varieties, 8640 kernels, RGB, hyperspectral, VIS/NIR, seed-level, high-resolution, non-destructive imaging, phenotyping	SVM	99.84%	non-destructive, automatic feature extraction, spatio-spectral, no manual features, hyperspectral imaging	specialized hardware, not publicly available, low reproducibility, limited accessibility	Chatnuntawe et al., (2018). [2]

Development of a Convolutional Neural Networks-Based Model to Classify the Rice Varieties	Smartphone Rice Variety Image Dataset (4-class)	4 rice varieties (Bhalum-5, Shahrang, Nagina-22, IR-64), smartphone images	5-layer Convolutional Neural Network (CNN)	91.0%	Simple model, uses widely accessible images	Limited classes; no microscopic detail; not tested across many varieties	Kaur, S. et al., (2023).. [3]
Enhancing the Classification Accuracy of Rice Varieties Using CNNs	Ipsala, Arborio, Basmati, Jasmine, Karacadag rice image dataset (Turkish rice varieties)	Hyperspectral images of hybrid indica paddy rice grains (number of classes not publicly available; private dataset)	CARS (Competitive Adaptive Reweighted Sampling) + CNN + Transformer	95.33%	High classification accuracy, Spectral features exploited for better differentiation	Limited to 17 rice varieties; may not generalize to other varieties, High-resolution images required, which may not be readily available, The study did not explore the impact of environmental factors on classification accuracy.	Tran Thi Kim et al., (2023). [4]
Identification of Hybrid Indica Paddy Rice Grain Varieties Based on Hyperspectral Imaging and Deep Learning	Hyperspectral images of hybrid indica paddy rice grains	hyperspectral images, hybrid indica rice, grain-level data, spectral bands, high-resolution, non-destructive imaging	CNN	95.33%	high accuracy, spectral features, enhanced differentiation, non-destructive	specialized hardware, not RGB-compatible, limited generalization, reduced accessibility	Meng Zhang. et al., (2024). [5]

An Overall Real-Time Mechanism for Classification and Quality Evaluation of Rice	Rice Grain Dataset	20,000 images of 6 rice varieties in China (private dataset); includes classification, completeness grading, and chalkiness evaluation	One-stage object detection + CNN + ML	97.89%	Integrated pipeline for classification and quality evaluation, Multi-task capability, high accuracy in both classification and detection	Large system complexity, unknown generalization to unseen rice varieties, requires substantial computational resources	Wanke Xia. et al.,(2025) .[6]
An Efficient Classification of Rice Variety with Quantized Neural Networks	Ipsala–Arborio–Basmati–Jasmine–Karacadag Rice Image Dataset	75,000 rice grain images (15,000 per variety) of 5 types: Ipsala, Arborio, Basmati, Jasmine, and Karacadag. Each image is 256×256 pixels.	Quantized Neural Networks (QNN) based on Multi-Layer Perceptron (MLP) and LeNet-5 architectures with varying quantization levels.	99.87%	Extremely low memory usage: only 23.1 KB for model parameters, 23× fewer billion operations per second (GOPs) compared to similar studies, Suitable for deployment on resource-constrained edge devices.	Quantization may reduce model precision, Model complexity trade-offs may affect performance.	Tasci, et al.(2023)[7]

Classification of Rice Texture Based on Rice Image Using Inception V3 CNN	Rice Texture Image Dataset (Private, Indonesia)	1,560 rice images, multiple classes	Inception V3 Convolutional Neural Network (CNN)	95.7%	High classification accuracy.Utilizes a modern CNN architecture.Suitable for texture-based classification	The dataset is small, which may lead to overfitting. Limited variety of rice textures in the dataset	Budiono, G. R. et al.,(2023) [8]
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Reference:

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